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Statistical Models

Week 3: Asymptotics, Likelihood Ratio Statistics, Delta Method and Generating Functions

We will finish the topic on likelihood theory with a proof of asymptotic normality of the maximum likelihood estimator, and a short discussion of likelihood ratio statistics. Very related to the use of asymptotic normality is the so called delta method that allows us to approximate the variance of a transformed random variable - this method will also be discussed.

In order to be prepared for the next big topic, generalized linear models, please recall the fundamentals of normal linear models. Exercises 3 will contain a problem about linear models.

Generalized linear models apply to random variables that have a distribution belonging to the so called exponential dispersion family. This class of distributions has some useful theoretical properties in common. Very handy tool to prove these properties (and a lot more) are the moment generating function and the related cumulant generating function. When we have finished likelihood theory, we will look at generating functions.

Relevant Readings

Davison, Sections 4.4.2 (asymptotics of mle), 4.5.1 and a bit of 4.5.2 (likelihood ratio statistic), 2.2.2 (Delta method), 2.4 (moment and cumulant generating functions) evt. 8.1 (linear models).

Lecture Material

- Link to onenotes on Google drive
- Confidence interval based on likelihood-ration statistic, Davison Example 4.29
- Experiment with the Delta Method

Exercises 3

(note that these exercises are first due in course week 4)

Problem 3.1: Properties of approximate confidence intervals

Which approximate confidence intervals are better: those based on normal approximation of the maximum likelihood estimator or those based on the likelihood ratio statistic? Compare these two for the case of the scale parameter of exponential distributed data, see this example from the lecture. Design a simulation experiment to investigate how true coverage and mean length of these confidence intervals depend on sample size n. Alternatively, you can also calculate the true coverage - use the fact that the sum of n independent exponential distributed r.v. with mean 1 has a Gamma(n) distribution.

Problem 3.2: Examples for the delta method for univariate random variables

Suppose that $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Gamma}(r, \lambda)$.

a. Find the variance of $\bar{X}$.
b. Use the delta method to find the approximate variance of $\overline{X}^2$.
c. Use the delta method to find the approximate variance of $\log(\overline{X})$.
d. Use the delta method to find the approximate variance of $\sqrt{\overline{X}}$.
e. Check your answers in a–d using simulation.

(see MSRR[1], Exercise 13.5)

Problem 3.3: Approximate confidence interval for a ratio

This is an exercise where you apply the delta method for two independent random variables. How to do that is described in Davison, p.34 f., in particular Example 2.25.
One application where the delta method is particularly helpful is when you need a good guess for the variance of a ratio (Davison, Example 2.25). An example where the ratio was interesting is known from the book that we used in the course "Mathematical Statistics", MSRR[1]: there, the suspicion was raised that the telecom giant Verizon delays the repair times for customers from competing local carriers (CLEC) in relation to own customers (ILEC). To express this, we are interested in the ratio of the mean repair times for CLEC and ILEC customers.
When dealing with ratios, it is often advised to look at the logarithm of the ratio instead. One reason is that this statistic is symmetric. In the current exercise, you consider both the ratio and the log ratio. We estimate the ratio of (population) means by the ratio of sample means. This estimator is not unbiased (why?), but the bias is small if the variance of the denominator is small.
Start with theoretical calculations:

a. Let $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_m$ be independent random variables with means μ_X and μ_Y and variances σ_X^2 and σ_Y^2, respectively.
Show that the delta method gives approximate variances

$$\text{Var}\left[\frac{\overline{X}}{\overline{Y}}\right] \approx \frac{\sigma_X^2}{n} \cdot \frac{1}{\mu_Y^2} + \frac{\sigma_Y^2}{m} \cdot \frac{\mu_X^2}{\mu_Y^4}$$

and

$$\text{Var}\left[\log\left(\frac{\overline{X}}{\overline{Y}}\right)\right] \approx \frac{\sigma_X^2}{n} \cdot \frac{1}{\mu_X^2} + \frac{\sigma_Y^2}{m} \cdot \frac{1}{\mu_Y^2}.$$

Then apply these results to the Verizon data. You can find them in the R-package `resampledata3` or alternatively download `Verizon.csv` directly or find it on the `data` folder on Google Drive.

b. Calculate the approximate means and variances of ratio and log ratio from the data, where X stands for the CLEC ($n = 23$ repair times) and Y for the ILEC (these were $m = 1664$ repair times). To this end replace theoretical mean and variance by the sample mean and variance of the data.

c. Use the means and variances from part (b) to calculate an approximate 95% confidence interval for the ratio of means, based on normal approximation. Transform the confidence interval for the log ratio into a confidence interval for the original ratio. Comment on your results.

d. (optional) Use bootstrap to estimate the variances of $\overline{X}/\overline{Y}$ and $\log(\overline{X}/\overline{Y})$ for the Verizon data. Compare with your results from the delta method in (b).

(Inspired by MSRR[1], Exercises 13.6, 13.7 and 13.8)

[1] Chihara and Hesterberg (2019) Mathematical Statistics with Resampling and R, 2nd edition, Wiley, Hoboken NJ.

The last exercise is meant as a refresher on normal linear models in preparation for the coming module on generalized linear models (glms). You can read more on normal linear models in Chapter 8 of Davison(2003). Sections 8.1 - 8.5 cover mainly the topic as taught from the course "Mathematical Statistics", see Chapter 11 of the course notes from 2024 (Davison gives slightly more advanced details). I suggest that you first read Exercise 4 and if you can solve it without major problems you are good to go for glms. Otherwise glance through the first sections of Chapter 8 in Davison, or Section 11.1 in the course notes. The course notes also contain useful R-codes and recipes for specifying model formulas in R. Find a brief overview in the extra onenote notes `review-linear-models.pdf`.

Problem 3.4: Interpretation of estimated parameters in an "analysis of covariance" model

For 72 young girls suffering from anorexia, the `Anorexia.csv` file that you can download from the `data` folder on Google Drive shows their weights before and after an experimental period. The girls were randomly assigned to receive one of three therapies during this period. A control group received the standard therapy, which was compared to family therapy and cognitive behavioral therapy. Download the data and fit a linear model relating the weight after the experimental period to the initial weight and the therapy.
For this exercise, you can assume an ordinary linear normal model. It is a good idea to make a suitable plot of the data before calculations, and to draw the regression lines. The situation calls for an "analysis of covariance" model: regression lines with group specific intercept and slope.
Additional question: Instead of considering the "after" weight as target variable, one might want to look at the difference "after" - "before" . What happens then with the estimates?
(adapted from Agresti (2015), Exercise 1.24).